

# NAB2 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes NAB2 gene fusions from the msk\_impact\_50k\_2026 study, covering 72 fusion events. 25/72 fusions (34.7%) were in-frame. 49/72 fusions (68.1%) retained the PF04905 domain. Domain retention was tested with Fisher's exact test ( $p=0.11258$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for NAB2.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 72 analyzed fusion events, identifying 14 distinct fusion partner genes. The most frequent partner was STAT6, observed in 53 events.

### Domain retention

PF04905 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 20/25 (80.0%) of in-frame fusions retained the domain,  $p=0.11258$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 1/25 (4.0%) of in-frame fusions disrupted the domain,  $p=0.998249$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

### Cutpoint detection

Cutpoint detection scanned 71 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 372 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is 6 aa from the nearest configured domain boundary at 366 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=49$ ) and "not\_retained" ( $n=22$ ). For Frame\_status, retained: 20/21 (95.2%, 95% CI 77.3%-99.2%); not\_retained: 5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 56.8776 vs 78.2727,  $p=0.354421$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (14 row(s) in composite\_evidence\_ranking); see the accompanying table.

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: NAB2 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| STAT6        | 53          | 0.7361111111111112   | 0.20043213737826426    | 0.0                     | 0.34366576819407013      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.3891339296891707  | 1    |
| CUX2         | 1           | 0.013888888888888888 | 0.20043213737826426    | 0.0                     | 0.948051948051948        | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2631251900011857  | 2    |
| PDZRN4       | 1           | 0.013888888888888888 | 0.20043213737826426    | 0.0                     | 0.948051948051948        | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2631251900011857  | 3    |
| MGAT4C       | 2           | 0.027777777777777776 | 0.20043213737826426    | 0.0                     | 0.9090909090909091       | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26144770082369656 | 4    |
| KRT6A        | 1           | 0.013888888888888888 | 0.20043213737826426    | 0.0                     | 0.935064935064935        | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2611771380531338  | 5    |
| LRP1         | 3           | 0.041666666666666664 | 0.20043213737826426    | 0.0                     | 0.8658008658008658       | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2591208609968567  | 6    |
| TMEM117      | 2           | 0.027777777777777776 | 0.20043213737826426    | 0.0                     | 0.6558441558441559       | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2234606878366836  | 7    |
| KIF5A        | 1           | 0.013888888888888888 | 0.20043213737826426    | 0.0                     | 0.6558441558441559       | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2192940211700169  | 8    |
| FMN1         | 3           | 0.041666666666666664 | 0.20043213737826426    | 0.0                     | 0.23376623376623373      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.16431566619166194 | 9    |

| Partner_gene | Event_count | Recurrence_score   | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|--------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| DLGAP2       | 1           | 0.0138888888888888 | 0.20043213737826426    | 0.0                     | 0.23376623376623373      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.15598233285832858 | 10   |
| FAM186A      | 1           | 0.0138888888888888 | 0.20043213737826426    | 0.0                     | 0.23376623376623373      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.15598233285832858 | 11   |
| LFNG         | 1           | 0.0138888888888888 | 0.20043213737826426    | 0.0                     | None                     | 0.6664269678216079         | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.14225576210987473 | 12   |
| LGR5         | 1           | 0.0138888888888888 | 0.20043213737826426    | 0.0                     | 0.09740259740259738      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.13552778740378313 | 13   |
| MTOR         | 1           | 0.0138888888888888 | 0.20043213737826426    | 0.0                     | 0.0                      | 0.6664269678216079         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.12091739779339353 | 14   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio           | p_value                | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|----------------------|------------------------|---------------------|
| 319      | 0                      | 49               | 4                      | 18               | 0.0                  | 0.00752854724253449    | 2.123288820003888   |
| 352      | 0                      | 49               | 5                      | 17               | 0.0                  | 0.0020225947815764307  | 2.6940911176014084  |
| 364      | 0                      | 49               | 13                     | 9                | 0.0                  | 8.560748975173139e-09  | 8.067488237477606   |
| 372      | 0                      | 49               | 14                     | 8                | 0.0                  | 1.3283920823544527e-09 | 8.876673721601218   |
| 381      | 3                      | 46               | 14                     | 8                | 0.037267080745341616 | 5.884988789596151e-07  | 6.230254360113864   |
| 382      | 3                      | 46               | 15                     | 7                | 0.030434782608695653 | 1.0398205008988075e-07 | 6.983041624352472   |
| 400      | 4                      | 45               | 15                     | 7                | 0.04148148148148148  | 4.330878206429079e-07  | 6.36342402938942    |
| 426      | 4                      | 45               | 16                     | 6                | 0.03333333333333333  | 7.233726445978198e-08  | 7.140637918749475   |
| 434      | 5                      | 44               | 16                     | 6                | 0.04261363636363636  | 2.7039639334463545e-07 | 6.567999105485465   |
| 490      | 45                     | 4                | 20                     | 2                | 1.125                | 1.0                    | -0.0                |
| 502      | 46                     | 3                | 20                     | 2                | 1.5333333333333334   | 0.6419888956213136     | 0.19247248378799386 |
| 508      | 47                     | 2                | 20                     | 2                | 2.35                 | 0.5828392348978783     | 0.23445122058408072 |
| 510      | 48                     | 1                | 20                     | 2                | 4.8                  | 0.2249846907532149     | 0.6478470327668787  |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio         | p_value            | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|--------------------|--------------------|--------------------|
| 511      | 48                     | 1                | 21                     | 1                | 2.2857142857142856 | 0.5267605633802818 | 0.2783867465185951 |
| 521      | 49                     | 0                | 21                     | 1                | None               | 0.3098591549295775 | 0.508835667896869  |

## domain\_disruption tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 1  | 12 |
| 24 | 34 |

*domain\_disruption\_descriptives*

| Domain_key              | Domain_name                              | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-------------------------|------------------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_disruption_pf04904 | NAB conserved region 1 (NCD1) SAM domain | 71                      | 58                   | 0               | 13               | 0.0                                             |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 20 | 29 |
| 5  | 17 |

*domain\_retention\_descriptives*

| Domain_key      | Domain_name                   | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-----------------|-------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_key_domain | NAB conserved region 2 (NCD2) | 71                      | 49                   | 13              | 9                | 0.3551986935220468                              |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 2    | 58                   | 72                | 0.8055555555555556 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| CUX2         | 1           |
| DLGAP2       | 1           |
| FAM186A      | 1           |
| FMN1         | 3           |
| KIF5A        | 1           |
| KRT6A        | 1           |
| LFNG         | 1           |
| LGR5         | 1           |
| LRP1         | 3           |
| MGAT4C       | 2           |
| MTOR         | 1           |
| PDZRN4       | 1           |
| STAT6        | 53          |
| TMEM117      | 2           |

## Per-event results (results.tsv)

| event_id                        | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_na<br>me  | part<br>ner_<br>gen<br>e | fra<br>me<br>_sta<br>tus | tar<br>get<br>_ro<br>le  | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains | lost<br>_do<br>mai<br>ns                                                                                                                   | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                           | source_site2<br>_effect_on_f<br>rame |
|---------------------------------|---------------------------|----------------------------|----------------------|--------------------------|--------------------------|--------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0052<br>994-T01-IM<br>6-1 | P-0052<br>994-T0<br>1-IM6 |                            | FAM1<br>86A-<br>NAB2 | FAM<br>186<br>A          | out-<br>of-fr<br>ame     | thr<br>ee_<br>_pr<br>ime | 574883<br>61               | 7                           | 490                                 | True                           | lost                      | 0.0                              | False                       |                          | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>);<br>NAB<br>cons<br>erve<br>d re<br>gion<br>1 (N<br>CD1<br>) SA<br>M d<br>oma<br>in |                           | FAM186A<br>(NM_001145475) - NAB2<br>(NM_005967)<br>rearrangement: c.193-1572<br>6:FAM186A_c.1469-34:NA<br>B2inv   Protein Fusion: out<br>of frame {FAM186A:NAB2} | NA                                   |

| event_id                        | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status         | target_role             | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                                                                                               | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|---------------------------------|---------------------------|------------|-------------------|--------------|----------------------|-------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0015<br>989-T01-IM<br>6-2 | P-0015<br>989-T0<br>1-IM6 |            | FMN<br>1-NA<br>B2 | FMN<br>1     | out-<br>of-fr<br>ame | thr<br>ee<br>_pr<br>ime | 574879<br>44       | 7               | 490                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>);<br>NAB<br>cons<br>erve<br>d re<br>gion<br>1 (N<br>CD1<br>) SA<br>M d<br>oma<br>in |                   | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>Rearrangement : t(12,15)(q<br>13.3;q13.3)(chr12:g.57487<br>944::chr15:g.33270869)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2} | NA                           |
| EVT-P-0015<br>989-T05-IM<br>6-3 | P-0015<br>989-T0<br>5-IM6 |            | FMN<br>1-NA<br>B2 | FMN<br>1     | out-<br>of-fr<br>ame | thr<br>ee<br>_pr<br>ime | 574879<br>44       | 7               | 490                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>);<br>NAB<br>cons<br>erve<br>d re<br>gion<br>1 (N<br>CD1<br>) SA<br>M d<br>oma<br>in |                   | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>rearrangement: t(12,15)(q1<br>3.3;q13.3)(chr12:g.574879<br>44::chr15:g.33270870)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2}  | NA                           |
| EVT-P-0015<br>989-T02-IM<br>6-4 | P-0015<br>989-T0<br>2-IM6 |            | FMN<br>1-NA<br>B2 | FMN<br>1     | out-<br>of-fr<br>ame | thr<br>ee<br>_pr<br>ime | 574879<br>45       | 7               | 490                         | True                   | lost          | 0.0                      | False               |                  | NAB<br>cons<br>erve<br>d re<br>gion<br>2 (N<br>CD2<br>);<br>NAB<br>cons<br>erve<br>d re<br>gion<br>1 (N<br>CD1<br>) SA<br>M d<br>oma<br>in |                   | FMN1 (NM_001103184) -<br>NAB2 (NM_005967)<br>Rearrangement : t(12,15)(q<br>13.3;q13.3)(chr12:g.57487<br>944::chr15:g.33270869)  <br>Protein Fusion: out of<br>frame {FMN1:NAB2} | NA                           |

| event_id                        | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains                                                            | disrupted_domains | source_annotation_text                                                                                                         | source_site2_effect_on_frame |
|---------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------|-------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0020<br>678-T01-IM<br>6-6 | P-0020<br>678-T01-IM6 |            | KRT6A-NAB2  | KRT6A        | in-frame     | three_prime | 57486817           | 5               | 382                         | True                   | lost          | 0.0                      | False               |                  | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                   | KRT6A (NM_005554) - NAB2 (NM_005967) rearrangement: c.1204-574:KRT6A_c.1144-29:NAB2inv   Protein Fusion: in frame {KRT6A:NAB2} | NA                           |
| EVT-P-0045<br>531-T01-IM<br>6-7 | P-0045<br>531-T01-IM6 |            | LGR5-NAB2   | LGR5         | unknown      | three_prime | 57488458           | 7               | 511                         | False                  | lost          | 0.0                      | False               |                  | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                   | LGR5 (NM_003667) - NAB2 (NM_005967) rearrangement: c.285-7692:LGR5_c.1532:NAB2dup   Protein Fusion: mid-exon {LGR5:NAB2}       | NA                           |
| EVT-P-0068<br>029-T01-IM<br>7-9 | P-0068<br>029-T01-IM7 |            | LRP1-NAB2   | LRP1         | unknown      | three_prime | 57487182           | 6               | 426                         | True                   | lost          | 0.0                      | False               |                  | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                   | LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.609:LRP1_c.1277-8:NAB2dup   Protein Fusion: mid-exon {LRP1:NAB2}          | NA                           |



| event_id                         | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                        | lost_domains                                                            | disrupted_domains             | source_annotation_text                                                                                                                                   | source_site2_effect_on_frame |
|----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0058<br>847-T01-IM<br>7-10 | P-0058<br>847-T01-IM7 |            | LRP1-NAB2   | LRP1         | unknown      | three_prime | 57486722           | 4               | 372                         | False                  | lost          | 0.0                      | False               |                                                                         | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                               | LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.5662+685:LRP1_c.1114:NAB2dup   Protein Fusion: mid-exon {LRP1:NAB2}                                 | NA                           |
| EVT-P-0008<br>384-T01-IM<br>5-12 | P-0008<br>384-T01-IM5 |            | NAB2-CUX2   | CUX2         | out-of-frame | three_prime | 57486406           | 4               | 364                         | True                   | disrupted     | 0.017964071856287425     | True                |                                                                         | NAB conserved region 1 (NCD1) SAM domain                                | NAB conserved region 2 (NCD2) | CUX2 (NM_015267) - NAB2 (NM_005967) rearrangement: c.63+83058:CUX2_c.1091+42:NAB2dup   Protein fusion: out of frame (CUX2-NAB2)                          | NA                           |
| EVT-P-0038<br>472-T01-IM<br>6-13 | P-0038<br>472-T01-IM6 |            | NAB2-DLGAP2 | DLGAP2       | out-of-frame | five_prime  | 57487547           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                                                                         |                               | NAB2 (NM_005967) - DLGAP2 (NM_004745) rearrangement: t(8;12)(p23.3;q13.3)(chr8:g.1518542::chr12:g.57487547)   Protein Fusion: out of frame {NAB2:DLGAP2} | NA                           |
| EVT-P-0039<br>139-T01-IM<br>6-14 | P-0039<br>139-T01-IM6 |            | NAB2-KIF5A  | KIF5A        | in-frame     | five_prime  | 57486074           | 2               | 319                         | True                   | disrupted     | 0.718562874251497        | True                | NAB conserved region 1 (NCD1) SAM domain                                |                                                                         | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - KIF5A (NM_004984) rearrangement: c.958-157:NAB2_c.129+5819:KIF5Adeletion   Protein Fusion: in frame {NAB2:KIF5A}                      | NA                           |

| event_id                         | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                                                            | disrupted_domains             | source_annotation_text                                                                                                                       | source_site2_effect_on_frame |
|----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|-------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022<br>192-T01-IM<br>6-18 | P-0022<br>192-T01-IM6 |            | NAB2-LRP1   | LRP1         | out-of-frame | three_prime | 57486554           | 4               | 364                         | True                   | disrupted     | 0.017964071856287425     | True                |                                          | NAB conserved region 1 (NCD1) SAM domain                                | NAB conserved region 2 (NCD2) | LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.842-2141:LRP1_c.1092-146:NAB2dup   Protein Fusion: out of frame {LRP1:NAB2}             | NA                           |
| EVT-P-0028<br>347-T01-IM<br>6-21 | P-0028<br>347-T01-IM6 |            | NAB2-MGAT4C | MGAT4C       | unknown      | five_prime  | 57486329           | 3               | 352                         | False                  | disrupted     | 0.9161676646706587       | True                | NAB conserved region 1 (NCD1) SAM domain |                                                                         | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) rearrangement: c.1056:NA B2_chr12:g.86871854inv   Protein Fusion: mid-exon {NAB2:MGAT4C}                                    | NA                           |
| EVT-P-0011<br>161-T01-IM<br>5-22 | P-0011<br>161-T01-IM5 |            | NAB2-MGAT4C | MGAT4C       | unknown      | three_prime | 57486534           | 4               | 364                         | True                   | disrupted     | 0.017964071856287425     | True                |                                          | NAB conserved region 1 (NCD1) SAM domain                                | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) rearrangement: c.1092-166:NAB2_chr12:g.87188994del   Antisense fusion                                                       | NA                           |
| EVT-P-0025<br>245-T01-IM<br>6-23 | P-0025<br>245-T01-IM6 |            | NAB2-MTOR   | MTOR         | unknown      | three_prime | 57491919           | 7               | 526                         | True                   | lost          | 0.0                      | False               |                                          | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                               | MTOR (NM_004958) - STAT6 (NM_003153) rearrangement: t(1;12)(p36.22;q13.3)(chr1:g.11300310::chr12:g.57491919)   Transcript Fusion {MTOR:NAB2} | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                       | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0001<br>055-T02-IM5-49 | P-0001<br>055-T02-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487938           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused with STAT6 exon16) : c.1469-457:NAB2_c.1760:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6)  | NA                           |
| EVT-P-0066<br>534-T01-IM7-50 | P-0066<br>534-T01-IM7 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488034           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-361:NAB2_c.1775:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                  | NA                           |
| EVT-P-0003<br>845-T01-IM5-51 | P-0003<br>845-T01-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487834           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) -STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 16) : c.1468+453:NAB2_c.1744+30:STAT6inv   Protein fusion: in frame (NAB2-STAT6) | NA                           |
| EVT-P-0029<br>101-T01-IM6-52 | P-0029<br>101-T01-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57486824           | 4               | 381                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1144-22:NAB2_c.-1269:STAT6inv   Transcript Fusion {NAB2:STAT6}    | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0029763-T01-IM6-53 | P-0029763-T01-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487587           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1468+206:NAB2_c.1624:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                | NA                           |
| EVT-P-0001055-T03-IM6-55 | P-0001055-T03-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487938           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457:NAB2_c.1760:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                     | NA                           |
| EVT-P-0020523-T01-IM6-56 | P-0020523-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487505           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+124:NAB2_c.1745-112:STAT6inv   Protein Fusion: in frame {NAB2:STAT6} | NA                           |
| EVT-P-0032069-T01-IM6-57 | P-0032069-T01-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487712           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+331:NAB2_c.1870:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                       | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022<br>259-T02-IM<br>6-61 | P-0022<br>259-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488455           | 7               | 510                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-7 fused with STAT6 exons 16-22): c.1529:NAB2_c.1745-128:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}           | NA                           |
| EVT-P-0004<br>282-T03-IM<br>5-62 | P-0004<br>282-T0<br>3-IM5 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57488115           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exon 16 to 22) : c.1469-280:NAB2_c.1745-157:STAT6inv   Protein fusion: in frame (NAB2-STAT6) | NA                           |
| EVT-P-0033<br>539-T02-IM<br>6-63 | P-0033<br>539-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57486902           | 5               | 400                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) rearrangement: c.1200:NA B2_chr12:g.57486902inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                                     | NA                           |
| EVT-P-0019<br>357-T03-IM<br>6-64 | P-0019<br>357-T0<br>3-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488027           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-368:NAB2_c.1690:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}             | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0019<br>182-T01-IM<br>6-66 | P-0019<br>182-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487665           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+284: NAB2_c.1744+39:STAT6inv   Protein Fusion: in frame {NAB2:STAT6} | NA                           |
| EVT-P-0036<br>288-T01-IM<br>6-67 | P-0036<br>288-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488128           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1469-267:NAB2_c.1850:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                            | NA                           |
| EVT-P-0036<br>975-T01-IM<br>6-68 | P-0036<br>975-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57486781           | 4               | 381                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) rearrangement: c.1143+30: NAB2_chr12:g.57508083inv   Transcript Fusion {NAB2:STAT6}                                                                            | NA                           |
| EVT-P-0019<br>032-T01-IM<br>6-69 | P-0019<br>032-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488347           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-48:NAB2_c.1691:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                 | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name       | part_ner_gene | fra_me_status   | tar_get_role       | breakp_o_int_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains                                                                                                    | lost_do_mains | disrupt_ed_do_mains                            | source_annotation_text                                                                                                                                                                     | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|--------------------|---------------|-----------------|--------------------|-----------------------|-------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0017<br>518-T01-IM<br>6-70 | P-0017<br>518-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574865<br>97          | 3                 | 364                          | True                     | disru<br>pted  | 0.9880239<br>520958084     | True                 | NAB c<br>onserv<br>ed<br>region<br>1 (NC<br>D1)<br>SAM d<br>omain                                                     |               | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-2<br>fused to STAT6 exons<br>18-22): c.1092-103:NAB2_<br>c.2042:STAT6inv   Protein<br>Fusion: mid-exon<br>{NAB2:STAT6}     | NA                            |
| EVT-P-0016<br>751-T02-IM<br>6-71 | P-0016<br>751-T0<br>2-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574879<br>58          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2);<br>NAB c<br>onserv<br>ed<br>region<br>1 (NC<br>D1)<br>SAM d<br>omain |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exons 1-6<br>fused to STAT6 exons<br>16-22): c.1469-437:NAB2_<br>c.1745-109:STAT6inv  <br>Protein Fusion: in frame<br>{NAB2:STAT6} | NA                            |
| EVT-P-0038<br>591-T01-IM<br>6-72 | P-0038<br>591-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | in-fr<br>ame    | five<br>_pr<br>ime | 574877<br>47          | 6                 | 490                          | True                     | retai<br>ned   | 1.0                        | False                | NAB c<br>onserv<br>ed<br>region<br>2 (NC<br>D2);<br>NAB c<br>onserv<br>ed<br>region<br>1 (NC<br>D1)<br>SAM d<br>omain |               |                                                | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>Fusion (NAB2 exon6 fused<br>with STAT6 exon16) : c.14<br>68+366:NAB2_c.1745-67:<br>STAT6inv   Protein Fusion:<br>in frame {NAB2:STAT6}          | NA                            |
| EVT-P-0016<br>113-T01-IM<br>6-73 | P-0016<br>113-T0<br>1-IM6 |              | NAB2<br>-STA<br>T6 | STA<br>T6     | unk<br>now<br>n | five<br>_pr<br>ime | 574864<br>82          | 3                 | 364                          | True                     | disru<br>pted  | 0.9880239<br>520958084     | True                 | NAB c<br>onserv<br>ed<br>region<br>1 (NC<br>D1)<br>SAM d<br>omain                                                     |               | NAB co<br>nserve<br>d<br>region<br>2<br>(NCD2) | NAB2 (NM_005967) -<br>STAT6 (NM_003153)<br>fusion (NAB2 exon3 fused<br>to STAT6 exon 18: c.1091<br>+118:NAB2_c.2051:STAT6<br>inv   Protein Fusion:<br>mid-exon {NAB2:STAT6}                | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0004<br>405-T01-IM<br>5-74 | P-0004<br>405-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57487215           | 6               | 434                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) -STAT6(NM_003153) Fusion ( NAB2 exon 6 fused to STAT6 exon 18) : c.1302:NAB2_c.1969:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6)                | NA                           |
| EVT-P-0004<br>943-T01-IM<br>5-75 | P-0004<br>943-T0<br>1-IM5 |            | NAB2-<br>STAT6 | STAT6        | unknown      | five_prime  | 57488487           | 7               | 521                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-7 with STAT6 exons 15-22) : c.1561:NAB2_c.1739:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |
| EVT-P-0040<br>364-T01-IM<br>6-76 | P-0040<br>364-T0<br>1-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487738           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                       | NA                           |
| EVT-P-0040<br>364-T02-IM<br>6-77 | P-0040<br>364-T0<br>2-IM6 |            | NAB2-<br>STAT6 | STAT6        | in-frame     | five_prime  | 57487738           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                       | NA                           |



| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0014<br>386-T01-IM6-78 | P-0014<br>386-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487858           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 17 - 22): c.1468+477:NAB2_c.1891+67:STAT6inv   Protein fusion: in frame (NAB2-STAT6) | NA                           |
| EVT-P-0043<br>579-T01-IM6-79 | P-0043<br>579-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488047           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-348:NAB2_c.1744+96:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0044<br>036-T01-IM6-80 | P-0044<br>036-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488037           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |
| EVT-P-0044<br>232-T02-IM6-81 | P-0044<br>232-T02-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488269           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                           | NA                           |

| event_id                         | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                       | source_site2_effect_on_frame |
|----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0001<br>947-T03-IM<br>6-82 | P-0001<br>947-T03-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487383           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2: NAB2_c.1769:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}   | NA                           |
| EVT-P-0045<br>590-T01-IM<br>6-83 | P-0045<br>590-T01-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487780           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399: NAB2_c.1789:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6} | NA                           |
| EVT-P-0046<br>398-T01-IM<br>6-84 | P-0046<br>398-T01-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487473           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}  | NA                           |
| EVT-P-0046<br>398-T02-IM<br>6-85 | P-0046<br>398-T02-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487473           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}  | NA                           |

| event_id                         | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains             | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0048<br>656-T01-IM<br>6-86 | P-0048<br>656-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57486417           | 3               | 364                         | True                   | disrupted     | 0.9880239520958084       | True                | NAB conserved region 1 (NC D1) SAM domain                                 |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1091+53: NAB2_c.1956-25:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                           | NA                           |
| EVT-P-0013<br>489-T01-IM<br>5-87 | P-0013<br>489-T01-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487506           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 15 to 22) : c.1468+125:NAB2_c.1625:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |
| EVT-P-0053<br>200-T02-IM<br>6-89 | P-0053<br>200-T02-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487609           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+228:NAB2_c.1891+45:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                           | NA                           |
| EVT-P-0013<br>261-T01-IM<br>5-90 | P-0013<br>261-T01-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487567           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 15): c.1468+186:NAB2_c.1703:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6)                | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                        | lost_domains | disrupted_domains             | source_annotation_text                                                                                                                                         | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------------------------------------------------|--------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0013<br>216-T02-IM<br>5-91 | P-0013<br>216-T0<br>2-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487806           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1468+425:NAB2_c.1745-142:STAT6inv   Protein fusion: in frame (NAB2-STAT6)                                       | NA                           |
| EVT-P-0055<br>865-T01-IM<br>6-92 | P-0055<br>865-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488325           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-70: NAB2_c.1744+36:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                 | NA                           |
| EVT-P-0055<br>909-T01-IM<br>6-93 | P-0055<br>909-T0<br>1-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57485966           | 2               | 319                         | True                   | disrupted     | 0.718562874251497        | True                | NAB conserved region 1 (NCD1) SAM domain                                |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.957+185: NAB2_c.340-42:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                  | NA                           |
| EVT-P-0012<br>841-T01-IM<br>5-94 | P-0012<br>841-T0<br>1-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487686           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 with STAT6 exons 15-22) : c.1468+305:NAB2_c.1712: STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                        | lost_domains | disrupted_domains             | source_annotation_text                                                                                                                                                           | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------------------------------------------------|--------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0057770-T01-IM6-95 | P-0057770-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57487467           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+86: NAB2_c.1744+3:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                                    | NA                           |
| EVT-P-0057786-T01-IM6-96 | P-0057786-T01-IM6 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488244           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-151: NAB2_c.1892-23:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                                  | NA                           |
| EVT-P-0012421-T01-IM5-97 | P-0012421-T01-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57486509           | 3               | 364                         | True                   | disrupted     | 0.9880239520958084       | True                | NAB conserved region 1 (NCD1) SAM domain                                |              | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-3 fused in-frame with STAT6 exons 20-22): c.1091+145: NAB2_c.2159+239:STAT6inv   Protein fusion: in frame (NAB2-STAT6) | NA                           |
| EVT-P-0007527-T01-IM5-98 | P-0007527-T01-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488111           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |              |                               | NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-6 with STAT6 exons 16-22) : c.1469-284: NAB2_c.1745-161:STAT6inv   Protein fusion: in frame (NAB2-STAT6)    | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0001<br>055-T04-IM6-99  | P-0001<br>055-T04-IM6 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487938           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457:NAB2_c.1760:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0059<br>117-T01-IM7-100 | P-0059<br>117-T01-IM7 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488448           | 7               | 508                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1522:NA B2_c.287:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                                  | NA                           |
| EVT-P-0059<br>275-T01-IM7-101 | P-0059<br>275-T01-IM7 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487481           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+100:NAB2_c.1801:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                              | NA                           |
| EVT-P-0006<br>897-T01-IM5-102 | P-0006<br>897-T01-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488251           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2(NM_005967) - STAT6(NM_003153) Fusion ( NAB2 exons 1 to 6 fused with STAT6 exons 16 to 22) : c.1469-144:NAB2_c.1811:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0006<br>243-T01-IM5-103 | P-0006<br>243-T01-IM5 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57488431           | 7               | 502                         | False                  | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2(NM_005967)-STAT6(NM_003153) Fusion ( NAB2 exons 1 to 7 fused with STAT6 exon 16 to 22) : c.1505:NAB2_c.1745-142:STAT6inv   Protein fusion: mid-exon (NAB2-STAT6) | NA                           |
| EVT-P-0005<br>578-T01-IM5-104 | P-0005<br>578-T01-IM5 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57486815           | 4               | 381                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon4 fused with STAT6 exon5) : c.1144-31:NAB2_c.340-81:STAT6inv   Protein fusion: in frame (NAB2-STAT6)            | NA                           |
| EVT-P-0067<br>841-T01-IM7-106 | P-0067<br>841-T01-IM7 |            | NAB2-STAT6  | STAT6        | unknown      | five_prime  | 57487573           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv   Protein Fusion: mid-exon {NAB2:STAT6}                                           | NA                           |
| EVT-P-0066<br>305-T01-IM7-107 | P-0066<br>305-T01-IM7 |            | NAB2-STAT6  | STAT6        | in-frame     | five_prime  | 57488138           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain |              |                   | NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-257:NAB2_c.1745-14:STAT6inv   Protein Fusion: in frame {NAB2:STAT6}                                        | NA                           |

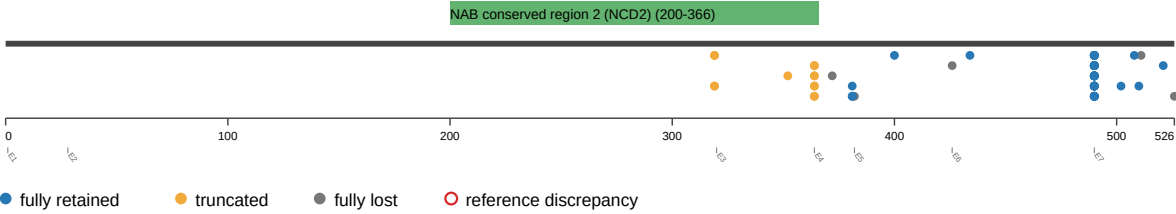
| event_id                      | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                        | lost_domains                             | disrupted_domains             | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0027<br>038-T01-IM6-108 | P-0027<br>038-T01-IM6 |            | NAB2-STAT6   | STAT6        | in-frame     | five_prime  | 57487452           | 6               | 490                         | True                   | retained      | 1.0                      | False               | NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain |                                          |                               | NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv   Protein Fusion: in frame {NAB2:STAT6} | NA                           |
| EVT-P-0030<br>371-T02-IM6-109 | P-0030<br>371-T02-IM6 |            | NAB2-TMEM117 | TMEM117      | out-of-frame | five_prime  | 57485885           | 2               | 319                         | True                   | disrupted     | 0.718562874251497        | True                | NAB conserved region 1 (NCD1) SAM domain                                |                                          | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - TMEM117 (NM_032256) rearrangement: c.957+104:NAB2_c.278-424:TMEM117dup   Protein Fusion: out of frame {NAB2:TMEM117}                            | NA                           |
| EVT-P-0004<br>638-T01-IM5-110 | P-0004<br>638-T01-IM5 |            | PDZRN4-NAB2  | PDZRN4       | out-of-frame | three_prime | 57486631           | 4               | 364                         | True                   | disrupted     | 0.017964071856287425     | True                |                                                                         | NAB conserved region 1 (NCD1) SAM domain | NAB conserved region 2 (NCD2) | PDZRN4 (NM_001164595) - NAB2 (NM_005967) - rearrangement: c.843+122859:PDZRN4_c.1092-69:NAB2del.   Protein fusion: out of frame (PDZRN4-NAB2)                      | NA                           |
| EVT-P-0030<br>371-T01-IM6-111 | P-0030<br>371-T01-IM6 |            | TMEM117-NAB2 | TMEM117      | out-of-frame | five_prime  | 57485885           | 2               | 319                         | True                   | disrupted     | 0.718562874251497        | True                | NAB conserved region 1 (NCD1) SAM domain                                |                                          | NAB conserved region 2 (NCD2) | NAB2 (NM_005967) - TMEM117 (NM_032256) Rearrangement: c.957+104:NAB2_c.278-424:TMEM117dup   Protein Fusion: out of frame {NAB2:TMEM117}                            | NA                           |



# Figures

## domain retention outliers

NAB2 domain-retention track



## fusion schematic

NAB2 fusion-transcript schematic

partner block → breakpoint → retained NAB2 portion (domain-colored, exon ticks)



## intragenic deletion schematic

NAB2 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk\_impact\_50k\_2026

